EXTRACTION AND REFINING OF METALS 14

In order to extract metals from their ores, energy is needed. The more reactive the metal, the more difficult it is to decompose (break down) its ore.

Copper The most important ores, the sulfides, contain between 1 and 12 percent of copper. Ores are mined and then crushed and concentrated by **flotation**. This technique allows comparatively heavy substances to float on water. Finely-ground copper ore is mixed with water and flotation reagents are added. Air is blown through this mixture to form a froth. The froth mixes with the sulfide but not with the other substances. The ore floats to the top, leaving other substances behind. Oxides and carbonates of copper are reduced (oxygen, etc. removed) with carbon. Sulfides are smelted in a furnace, resulting in copper that is approximately 98 percent pure. This can be further purified to 99.9 percent through electrolysis. Electrolysis uses large amounts of electricity to attract pure copper to an electrode.

1 Define each term.

a energy b sulfide c flotation d reagent e reactive

2 Draw a flow chart to show the production of copper metal from its ore.

Iron is extracted from iron ore (haematite) in a blast furnace. Coke (a form of carbon) and limestone (calcium carbonate) are added with iron ore to a blast furnace where the raw materials are strongly heated. The coke provides the heat to produce carbon monoxide (CO) which takes the oxygen from the iron oxide. The limestone removes impurities, which melt in the furnace and are removed as slag. Hot air is blown in near the bottom of the furnace. The molten iron produced is then cast into ingots. This cast iron usually requires further treatment before it can be used.

3 Suggest why it is important to remove impurities from substances being mined.

4 Draw a flow chart to show the production of iron metal from its ore.

More active metals such as aluminium (Al) need to be extracted from their ores by electrolysis. Sodium is extracted from sodium chloride (NaCl, rock salt) by this method.

5 Write balanced formula equations for each reaction.

a copper (I) sulfide + oxygen → copper + sulfur dioxide

b iron (III) oxide + carbon monoxide → iron + carbon dioxide

c sodium chloride → sodium + chlorine

d aluminium oxide → aluminium + oxygen

The Bronze Age This was the period of ancient human culture 4000–3000 BC. It was characterised by the human use of bronze. This age ended when the Iron Age began. The Greeks and Roman used bronze to make tools, weapons, coins and objects of art. Bronze is an alloy, usually containing copper and tin.

6 Define *alloy*.

The Iron Age This was a period in human culture when iron was commonly used for making tools and weapons. (Note that bronze was replaced by iron at different times in different cultures.) The first known use of iron was in Egypt (c. 4000 BC). Modern iron processing began in Europe in the 14th century.

7 Discuss why bronze was used by humans before iron was.

8 Match each term to the correct definition below:
alloy, electrolysis, furnace, haematite, ingot, ore, reduction, slag, smelt

a Heating an ore with a chemical change to obtain the metal: _____

b A piece of metal cast into a shape: _____

c A waste product of steel production: _____

d A mixture of metals: _____

e Passing an electrical current through a substance to cause a chemical reaction: _____

f A hard, red oxide of iron: _____

g A structure that uses heat to extract metals from their ores: _____

h The chemical removal of oxygen from a substance: _____

i A mineral that contains enough of a metal to be worth mining: _____

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